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Truncated Tetrahedron

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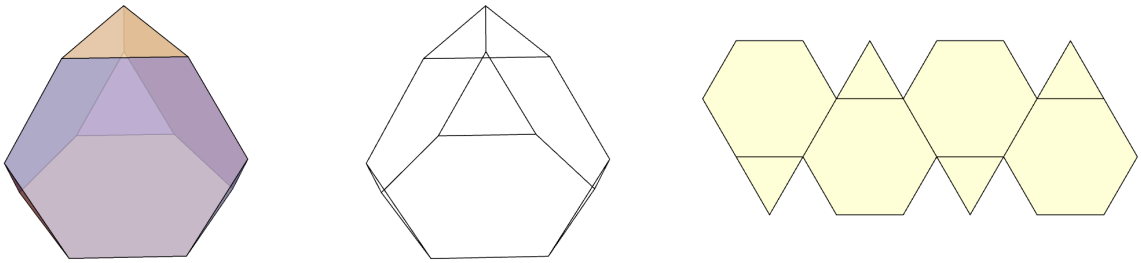

MAKE YOUR OWN TRUNCATED TETRAHEDRON

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The truncated tetrahedron is the Archimedean solid with faces $4\{3\}+4\{6\}$. It is also the uniform polyhedron with Maeder index 2 (Maeder 1997), Wenninger index 6 (Wenninger 1989), Coxeter index 16 (Coxeter *et al.* 1954), and Har'El index 7 (Har'El 1993). It has Schläfli symbol $t\{3,3\}$ and Wythoff symbol $23\mid 3$. It is illustrated above together with a wireframe version and a net that can be used for its construction.

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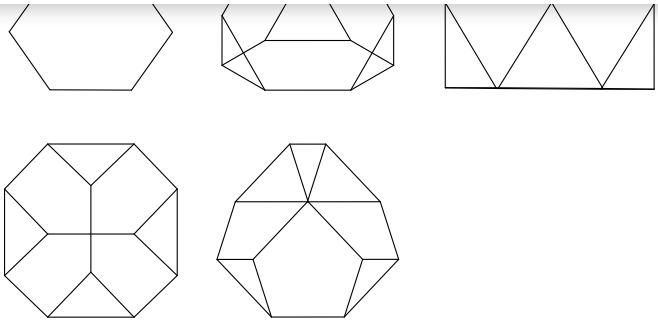
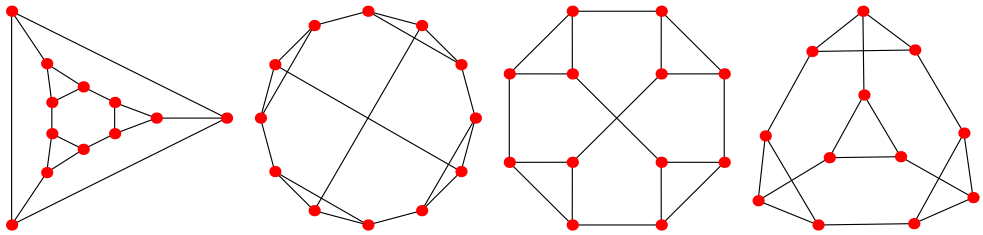
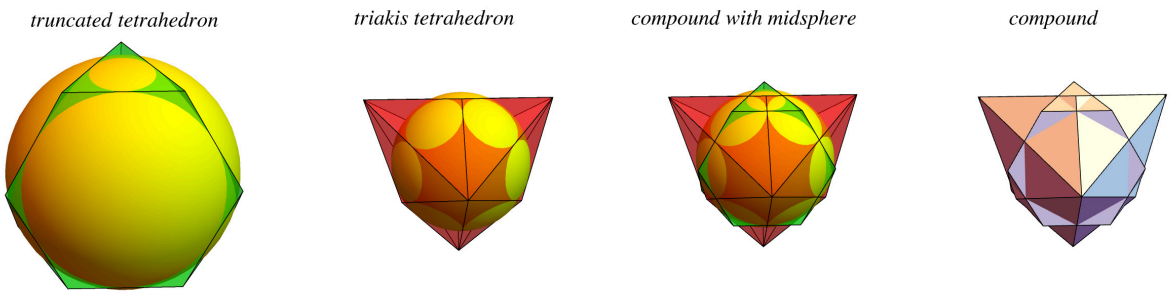


Some symmetric projections of the truncated tetrahedron are illustrated above.

It is implemented in the Wolfram Language as PolyhedronData["TruncatedTetrahedron"] or UniformPolyhedron["TruncatedTetrahedron"]. Precomputed properties are available as PolyhedronData["TruncatedTetrahedron"].



The skeleton of the truncated tetrahedron is the truncated tetrahedral graph, illustrated above in a number of embeddings.



The dual polyhedron of the truncated tetrahedron is the triakis tetrahedron, both of which are illustrated above together with their common midsphere. The inradius r of the dual, midradius ρ of the solid and dual, and circumradius R of the solid for $a = 1$ are

$$r = \frac{9}{44}\sqrt{22} \approx 0.95940 \tag{1}$$

$$\rho = \frac{3}{4}\sqrt{2} \approx 1.06066 \tag{2}$$

$$R = \frac{3}{4}\sqrt{22} \approx 1.17260. \tag{3}$$

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$$r_3 = \frac{5}{12}\sqrt{6} \tag{4}$$

$$r_6 = \frac{1}{4}\sqrt{6}\,. \tag{5}$$

The surface area and volume are

$$S = 7\sqrt{3} \tag{6}$$

$$V = \frac{23}{12}\sqrt{2}\,. \tag{7}$$

The unit truncated tetrahedron has Dehn invariant

$$D = 12\,\langle 3\rangle_2 \tag{8}$$

$$= 12\tan^{-1}\left(\sqrt{2}\right), \tag{9}$$

where the first expression uses the basis of Conway *et al.* (1999).

SEE ALSO

Archimedean Solid, Equilateral Zonohedron, Jabulani Polyhedron, Triakis Tetrahedron, Truncated Tetrahedron-Triakis Tetrahedron Compound

EXPLORE WITH WOLFRAM|ALPHA

WolframAlpha truncated tetrahedron

More things to try: = truncated tetrahedron = {25, 35, 10, 17, 29, 14, 21, 31} = five sixteenths

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